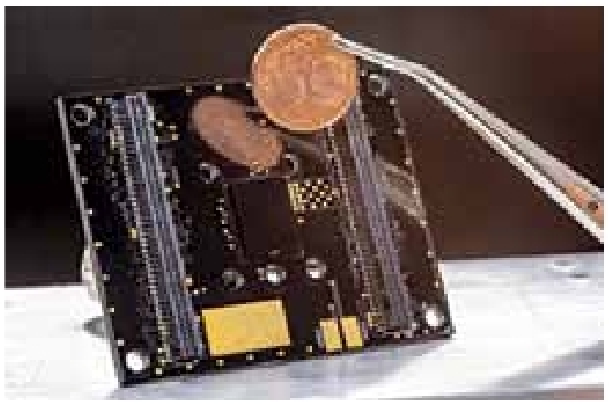
Sensor for air quality monitoring can fit in phones and wearables

A particle sensor developed by TU Graz, ams and Silicon Austria Labs is slightly smaller than two one-cent coins stacked on top of each other. Compact and energy-efficient, the 12 × 9 × 3mm innovation requires no maintenance and can be integrated into mobile devices such as smartphones, smartwatches, or fitness wristbands for measuring air quality in real-time and create an alert in the event of increased fine dust levels.

Besides wearables, the sensor can be integrated with local applications—both in the home and outdoors—and thus provide an unprecedented variety of measured values.



Prototype of the particle sensor (black square in the middle of the board) developed at TU Graz compared in size with a one-cent coin (Credit: Lunghammer - TU Graz)

Methanol fuel gives this tiny beetle bot the freedom to roam

A new robot beetle has been developed that can go a long-range distance on its own thanks to a methanol-fueled micro muscle (that can pack in 10x energy as tiny batteries). This can possibly lead to swarms of robotic insects that could assist search-and-rescue operations - even through tight spaces that are out of reach.

Look ma, no wires! A robotic beetle, which weighs about as much as three grains of rice, crawls, carries and climbs without being tethered to a power source (Credit: X. Yang, L. Chang, N.O. Pérez-Arancibia/Sci. Robotics 2020)

To turn methanol into motion, the researchers coated a nickel-titanium alloy wire with platinum. The alloy contracts like a muscle when heated and extends once cool. The platinum generates heat by combusting any methanol vapour that comes in contact with it. By varying the exposure to fuel in a periodic pattern, the temperature varies and the micromuscle accordions. That motion causes the bot's forelegs to rear up. When the legs move back again, the body drags forward.

The beetle bot weighs about as much as three grains of rice, on par with live insects, and crawls on flat surfaces while carrying up to 2.6 times its weight.

A smartwatch app alerts hard of hearing to nearby sounds

A new smartwatch app alerts users who are deaf or hard of hearing of nearby sounds such as microwave beeps or car horns. It can also be used for hearing urgent noises like fire alarm, door knock, or alarm clock.

Although sound awareness apps for smartphones exist, people who are deaf or hard of hearing generally prefer to wear such a device on the wrist rather than keep it inside a pocket or bag. The SoundWatch app pairs with an Android smartwatch and phone.

The watch records ambient noises and sends that data to the phone for processing. Up to twenty sounds can be identified with 81 to 97 per cent accuracy.

The new smartwatch app, SoundWatch, alerts people who are deaf and hard of hearing to bird songs, sirens and other noises (Credit: Jain et al./ASSETS 2020)

Cameras can now capture colours invisible to the human eye

Now a breakthrough research allows cameras to recognise colours that the human eye and even ordinary cameras are unable to perceive.

The technology makes it possible to visualise gases and substances such as hydrogen, carbon, and sodium—each of which has a unique colour in the infrared spectrum—as well as biological compounds that are found in nature but are invisible to the naked eye or ordinary cameras. It has groundbreaking applications in a variety of fields from computer gaming and photography to the disciplines of security, medicine, and astronomy.

Researchers were able to develop cheap and efficient technology that could be used with a standard camera for the conversion of photons of light from the entire mid-infrared region to the visible region, at frequencies that the human eye and the standard camera can pick up.